

1 Competition: Autarkies (7(+7) Points)

Write an anytime algorithm that computes the maximal-size autarky (in terms of the number of satisfied clauses) for a given CNF formula. The algorithm should output the set of literals representing the autarky as soon as possible, followed by improved sets iteratively. Use the specified SAT competition output format for partial (or complete) solutions. Each solution should be output on a single line starting with “v ” followed by literals separated by spaces. After k minutes, we will interrupt your algorithm and check the last output line, which should represent the best autarky yet. We will test its quality by computing the number of clauses satisfied by the autarky.

2 Variable Elimination (3 Points)

For a gate encoding E with output x in a formula $F = E \cup R$, we simplified the resolvents $(E_x \cup R_x) \otimes (E_{\bar{x}} \cup R_{\bar{x}})$ by $S := (E_x \otimes R_{\bar{x}}) \cup (R_x \otimes E_{\bar{x}})$, dropping both $R_x \otimes R_{\bar{x}}$ and $E_x \otimes E_{\bar{x}}$. Show that the clauses in $R_x \otimes R_{\bar{x}}$ can be derived from S by resolution. You can assume that E encodes a binary AND gate.

3 Variable Elimination (2×2 Points)

Let the formula S with gate encodings E_1 and E_2 be given. Apply variable elimination for gates for variables a and r . Give the clause sets after each elimination step. Try the following two strategies.

1. Eliminate variable a first, and then r if possible.
2. Eliminate variable r first, and then a if possible.

$$S = \underbrace{\{\neg x, \neg y, a\}, \{x, \neg a\}, \{y, \neg a\}}_{E_1}, \underbrace{\{\neg a, r\}, \{\neg z, r\}, \{a, z, \neg r\}, \{a, z, r\}, \{\neg a, \neg r\}}_{E_2}$$

4 Blocked Clauses (3×3 Points)

If Blocked Clause Elimination (BCE) reduces a formula F to the empty formula then F is called a blocked set. Prove the following statements.

1. Any formula F can be partitioned into two blocked sets S and L such that $F = S \cup L$. Design a linear algorithm that produces L and S from F .
2. Blocked sets are not closed under resolution. If F is a blocked set then $F \cup C_1 \otimes C_2$, where $C_1, C_2 \in F$ may not be a blocked set anymore.
3. Blocked sets are not closed under partially assigning variables. If F is a blocked set then $F_{x=v}$ (the result of assigning v to x and subsequent simplification) may not be a blocked set anymore.